

Shout out to Maciej Gorywoda who helped make this roadmap.

LinkedIn Profile

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roadmap.sh

Related Roadmaps

☒ Backend Roadmap

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Pick your IDE

IntelliJ IDEA

VS Code

Secondary Options

Vim

Emacs

Zed

Sublime

Class

Trait

Object

Functions & Methods

Functions & Methods

Method Calls

Anonymous Func / Lambda

Total/Partial Funcs

Funcs returning Funcs

Regex

Working with Strings

What different items mean

A concept that connects multiple language features, helping you how they work together.

Links related concepts that are complementary or alternative ways of doing the same thing.

You don't need to learn advanced topics linearly, pick them as you explore libraries and frameworks

private / protected

package

Implicit Parameter

Tail Recursion

Trampolines

Scala-centered

ScalaCLI

sbt

Mill

Common to JVM

Gradle

Maven

Pure Functions

Capture Checking

Unit Testing

ScalaTest

JUnit

mUnit

uTest

ZIO Test

specs2

Integration / Performance Testing

ScalaTest

Gatling

Cats Effect

ZIO

Ox

Gears

FS2

Kyo

Streaming

Akka & Peko Streams

ZIO Streams

Spark

Akka

Pekko

Indigo

CosPlay

LibGDX w/ Scala

Monocle

Chimney

Shapeless

Magnolia

Scalameta

Scala

Introduction

Setting Up Scala

Learn the Basics

Variables & Constants

Operators

Conditionals

Data Structures

Loops

Try/Catch

Case Classes

Sealed Traits

apply Method

Enums

Case Objects

unapply Method

Pattern Matching

Error Handling

Laziness

Scope & Visibility

Recursion

Tools, Ecosystem & Platforms

Build Tools

Ecosystems

Platforms

Advanced Topics

Referential Transparency

Monads

Capabilities

Typeclasses

Type System

Macros

Frameworks and Libraries

Testing

Backend

Effect Systems

GUI

Concurrency

Distributed Computing

Data Handling

Video game Engines

Macros & the Type System

Find Resources

Docs

Books

Courses

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Main Data Types

Integers

Float

Strings

Booleans

Unit

Nothing

Generic Types

Type Hierarchy

Type Parameters

Conversions

Collections

Array

Range

Seq

List

Vector

Set

Map

Mutable Collections

Common Collection Methods

find

filter

foreach

map

flatMap

foldLeft

collect / collectFirst

for comprehension

Option

Either

Try

Lazy vals

by-name Parameters

Lazy Collections

Iterators

Views

LazyList

Early returns

Cats

ZIO

Li Haoyi

Akka / Pekko

No Ecosystem

JVM

Scala Native

Scala.js

Category Theory

Variance

Context Bounds

sttp

Scalatra

Play

Tapir

AkkaHTTP

http4s

Web

Laminar

Calico

Slinky

React.js

Desktop

ScalaFX

JavaFX

Android

scala-android-plugin

Scala on Android

GraalVM Native Image

Database

Doobie

ScalikeJDBC

Quill

Slick

JSON

PlayJSON

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Jsoniter

uPickle

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MLOps

Data Engineer